

A Comparative Network and Sentiment Analysis of Rhythm Game Communities on YouTube

By: UG Group 59

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I. Introduction

Online fandom communities are an important part of social media culture, especially for games that rely on music, characters, events and ongoing fan participation. Rhythm games such as *Project SEKAI: Colorful Stage!* and *BanG Dream! Girls Band Party!* have active YouTube communities where users comment on videos, discuss songs and characters, react to updates and interact with other fans. These interactions can reveal how different fandoms organise themselves around shared interests whether the discussion patterns differ between similar communities on the same platform.

YouTube, as a platform, is useful for this analysis because it combines textual comments with interaction structures such as comment replies. This makes it possible to study both the emotional tone of discussion through sentiment analysis and the structure of user interaction through network analysis. Since both Project SEKAI and BanG Dream are rhythm game communities with similar music-centred fandom cultures, comparing them allows the project to focus on the differences in community behaviour rather than the differences between unrelated topics.

This report investigates how interaction networks and sentiment patterns differ between the Project SEKAI and BanG Dream YouTube communities. The analysis uses YouTube comment data collected from 30 videos for each game. Sentiment analysis was used to compare the emotional tone of comments, while network analysis was used to examine reply structures, user centrality, community clustering and modularity. Two sentiment approaches, VADER and RoBERTa, were included to compare whether the overall sentiment trends remained consistent across both a rule-based model and a contextual transformer-based model. Together, these methods help identify whether the two rhythm game fandoms show different forms of interaction and whether their communities appear clustered or self-reinforcing.

II. Research Question

The research question for this project is:

How do interaction network structures and sentiment patterns differ between Project SEKAI and BanG Dream rhythm game communities on YouTube?

a. Aim

The aim of this project is to compare two major rhythm game fandoms on YouTube using both text-based and network-based analysis. Specifically, the project examines:

- Whether the two fandoms differ in overall sentiment
- Whether VADER and RoBERTa identify similar or different sentiment patterns
- Whether their reply networks differ in size, density, clustering and modularity

- Whether highly central users show different sentiment patterns from peripheral users
- Whether detected network communities suggest clustered or self-reinforcing interaction patterns.

This question is suitable for social media and network analysis because YouTube comments provide both textual content and relational interaction data. The text allows sentiment analysis to be performed, while comment replies allow a user interaction network to be constructed. By comparing two similar rhythm game fandoms on the same platform, the project can focus on how community behaviour differs between fandoms while keeping the platform consistent.

III. Dataset Overview

The dataset contains YouTube comments collected from videos related to two rhythm game communities: Project SEKAI and BanG Dream. For each game, 30 YouTube videos were used. The final cleaned dataset contains a similar number of comments for both communities, making the comparison reasonably balanced.

Game	Raw Comments/Replies	Removed Empty Text	Removed Duplicates	Final Comments	Top-Level Comments	Replies	Unique Users	Final Reply Edges	Videos
Project SEKAI	13253	1	49	13203	8617	4586	9537	3372	30
BanG Dream	12977	8	21	12948	9621	3327	8290	2582	30

The Project SEKAI dataset contains slightly more comments, replies, unique users and reply edges than BanG Dream dataset. This suggests that the Project SEKAI videos in the sample generated more direct user interaction, while the BanG Dream dataset had a higher proportion of top-level comments compared with replies.

The data supports both required parts of the assignment. The comment text supports the sentiment and textual analysis, while the reply relationships between users support the network construction and analysis. The cleaned comments were used for sentiment analysis, and the reply edges were used to build separate interaction networks for each game.

IV. Success Criteria

The analysis is considered successful if it can produce meaningful evidence for comparing the two rhythm game communities. The main success criteria are:

1. Constructing a comparable YouTube reply networks for Project SEKAI and BanG Dream.
2. Calculating and comparing network measures such as nodes, edges, density, connected components, communities and modularity.

3. Performing sentiment analysis on comments from both communities using VADER and RoBERTa.
4. Comparing sentiment patterns between the two fandoms.
5. Comparing agreement and differences between the two sentiment models.
6. Examining whether central users differ from peripheral users in sentiment.
7. Interpreting whether the results suggest clustered or self-reinforcing interaction patterns.

The project does not aim to prove the existence of echo chambers in a strict causal sense. Rather, it investigates whether the YouTube reply networks will show any evidence of clustered interaction patterns and whether those clusters relate to the differences in sentiments across the two fandoms.

V. Data Collection

Data collection for this project utilised YouTube Data API v3, providing access to publicly available video metadata, comment threads, and user interaction data. These communities were selected as both are music-centred rhythm games with active YouTube fandom discussion, making them suitable for comparing sentiment and user interaction patterns.

Multiple search queries were used for each game to capture a broad range of content. Project SEKAI queries included terms such as “Project Sekai update”, “Project Sekai tier list”, and “Project Sekai community”. BanG Dream queries included “Bandori”, “Bandori gameplay”, and “Girls Band Party”. This helped collect gameplay, update opinion, and community-related videos.

The YouTube search endpoint was used to retrieve candidate videos, with duplicate videos removed across queries. Up to 30 videos per game were retained. For each video, metadata including title, publication date, view count, like count, and comment count were collected.

Top-level comments were collected using the “commentThreads” endpoint, with up to 500 top-level comments per video. Videos with fewer than 100 total comments or replies were excluded. Replies were also collected using the “comments” endpoint, with 20 replies per thread, allowing user reply relationships to be used for network construction.

For each comment and reply, the dataset stored both text fields for sentiment analysis and relational fields for network analysis, including “authorChannelId”, “parentCommentId”, and “commentId”.

The final output was stored as structured JSON files containing video metadata and nested comment/reply data for each game.

VI. Data Preprocessing

The raw YouTube API output was stored as nested JSON files containing video-level metadata and comment/reply records. To prepare for the data analysis, the JSON structure was first flattened into tabular format, with video metadata merged into each comment row. This produced a structured dataset where each row represented a single top-level comment or reply.

Comment text was then cleaned to improve consistency for sentiment analysis. HTML entities were decoded, URLs were removed, and excessive whitespace was normalised. Comments with empty text after cleaning were excluded from the dataset. Duplicate records were also removed based on game label, comment identifier, and cleaned text.

To prepare the network analysis dataset, reply comments were converted into directed user-to-user edges using YouTube channel IDs. In this network, the source node represented the replying user and the target node represented the author of the parent comment. Repeated reply interactions between the same pair of users were aggregated into weighted edges, allowing the final network to capture both interaction structure and reply frequency.

The preprocessing stage produced two main outputs including a cleaned comment dataset for sentiment analysis, and an edge list dataset for network analysis.

VII. Sentiment Analysis Method

Sentiment analysis was performed using two approaches including VADER and RoBERTa. VADER was used as a rule-based sentiment model designed for short social media text, while RoBERTa was used as a transformer-based model that can account for more contextual language patterns. Using both models allowed the analysis to compare whether sentiment trends were consistent across different methods.

VADER generated positive, neutral, negative, and compound sentiment scores for each cleaned comment. The compound score ranged from -1 to 1 and was converted into sentiment categories using standard thresholds:

- Positive: compound score ≥ 0.05
- Negative: compound score ≤ -0.05
- Neutral: compound score between -0.05 and 0.05

RoBERTa used the cardiffnlp/twitter-roberta-base-sentiment-latest model to classify each cleaned comment as positive, neutral, or negative. Unlike VADER, RoBERTa considers the surrounding context of a comment, making it useful for informal fandom language, sarcasm, and ambiguous phrasing.

The two model outputs were then compared using sentiment distributions and model agreement rates. This helped identify whether differences between Project SEKAI and BanG Dream were consistent across both models, and whether either fandom contained comments that were more difficult to classify consistently.

VIII. Network Construction and Metrics

a. Nodes and Edges

A directed reply network was constructed for each game community. Each node represents a unique YouTube user identified by “authorChannelId”, and each directed edge represents one user replying to another user. Edges were weighted by the amount of repeated replies between the same pair of users.

b. Density and Connected Components

To compare the overall structure of the two networks, graph-level metrics were calculated including nodes, edges, density, connected components, largest connected component, average degree, clustering coefficient, and reciprocity. These metrics were used to assess how connected, fragmented, and interactive each fandom network was.

c. Community Detection and Modularity

For community detection, the directed reply networks were converted into undirected weighted graphs. The greedy modularity community detection algorithm was then applied to identify groups of users who interacted more frequently with each other than with the rest of the network. Modularity was used to measure how strongly separated these communities were.

d. Centrality

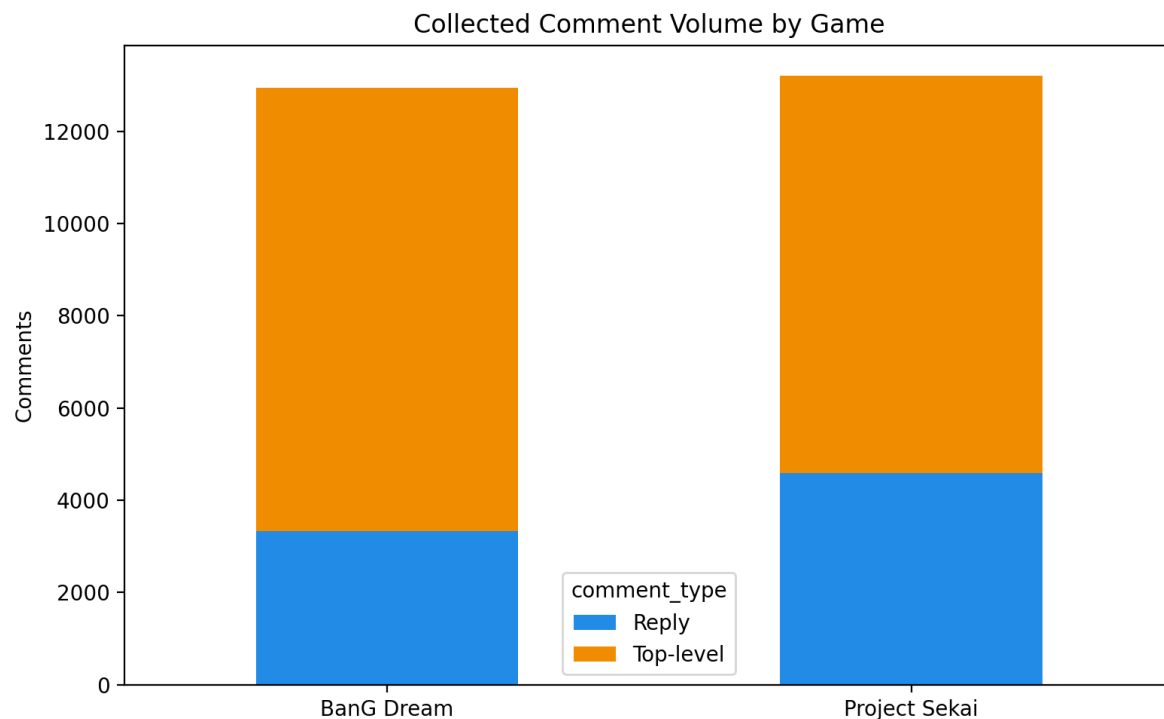
Node-level centrality measures were calculated to identify influential users in each network. Weighted degree measures overall interaction volume, PageRank measured importance based on network position, and betweenness centrality identified users who acted as bridges between parts of the network. These measures were later used to compare sentiment patterns between highly central and peripheral users.

IX. Result

This section will present the main findings from the sentiment and network analyses. The results compare the Project SEKAI and BanG Dream YouTube communities in terms of dataset size, sentiment distribution, reply network structure, centrality-based sentiment patterns and community clustering.

a. Final Dataset Overview

The final cleaned dataset contained a similar number of comments for both rhythm game communities. Project SEKAI had 13203 comments after preprocessing, while BanG Dream has 12,948 comments. Both communities were collected from 30 YouTube videos, making the dataset reasonably balanced for comparison.

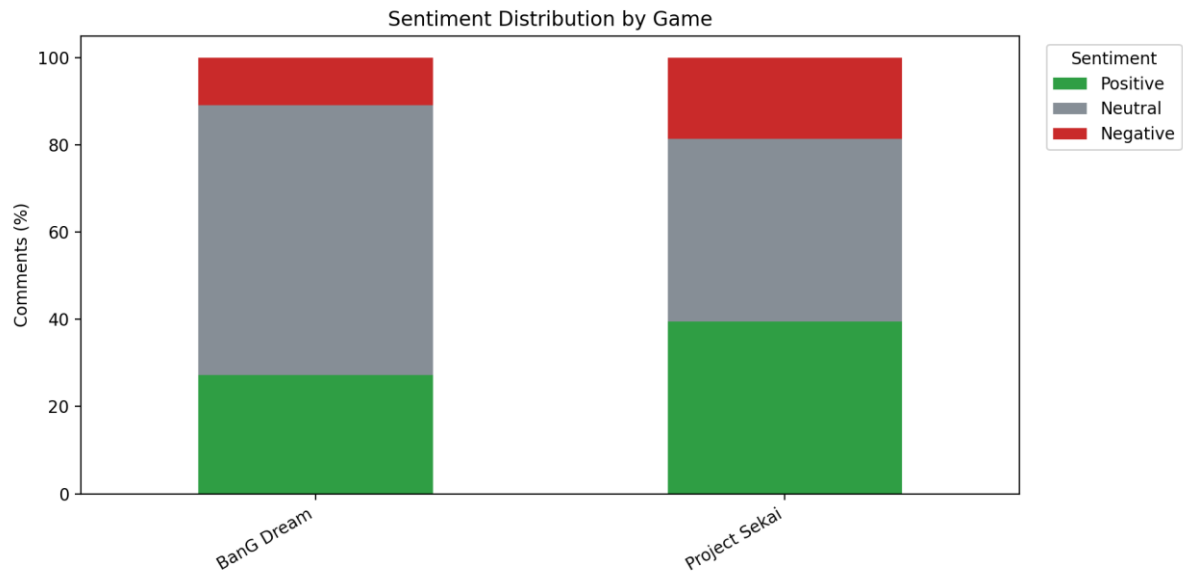


Project SEKAI had more unique users, replies and final reply edges than BanG Dream. This suggests that the Project SEKAI sample generated more direct reply-based interaction, while BanG Dream had a larger proportion of top-level comments. BanG Dream had 9621 top-level comments compared to their 3327 replies, while Project SEKAI shared 8617 top-level comments with 4586 replies.

This difference is important to note, as the network analysis is based on reply interactions. Thereby, Project SEKAI having more replies would mean its network has more direct user-to-user interaction available for analysis.

b. Sentiment Distribution

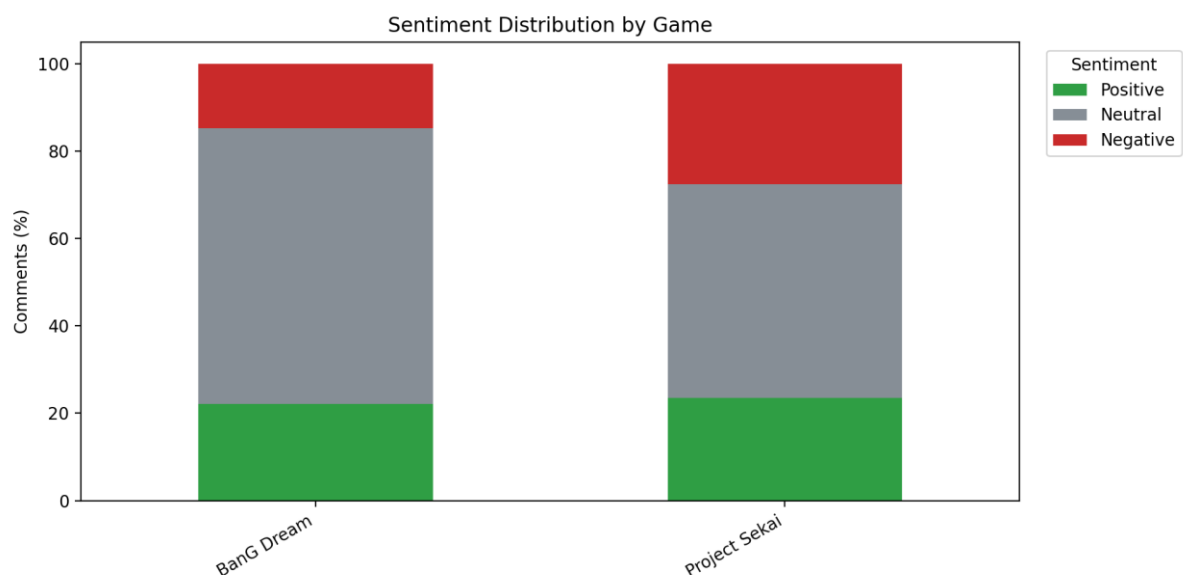
The sentiment distribution shows clear differences between the two rhythm game communities. Since both VADER and RoBERTa were used, the analysis compares whether the same broad trends appear across both models.



VADER

Using VADER, Project SEKAI had a higher percentage of both positive and negative comment sentiments. Positive comments made up 39.50% of their respective dataset, compared to the 27.23% of the same for BanG Dream. Negative comments were also more common within Project SEKAI, making up 18.61% of comments compared with 10.84% within BanG Dream.

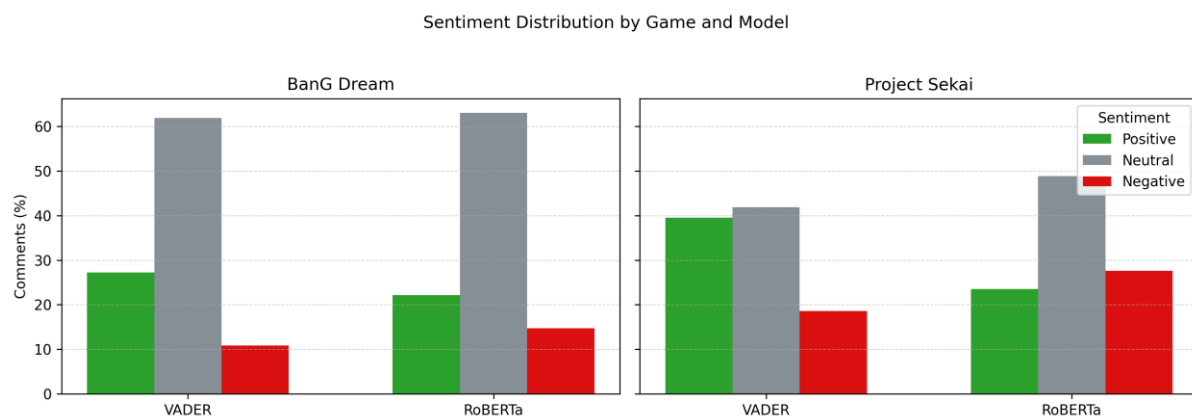
However, BanG Dream had a much higher neutral percentage, with 61.92% of its comments being identified and classified as neutral. In comparison, for Project SEKAI, only 41.89% of the comments were neutral.



RoBERTA

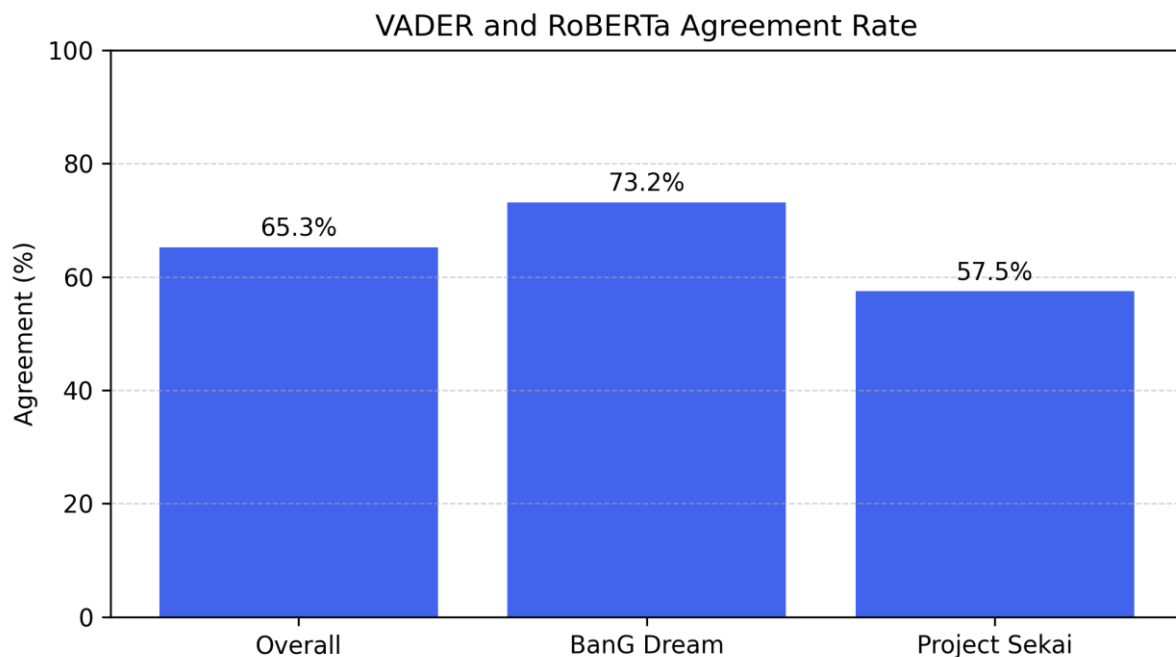
The RoBERTa results showed a similar overall pattern in terms of BanG Dream being more neutral, but it classified sentiment more cautiously for Project SEKAI. Under RoBERTa, Project SEKAI had 23.55% positive comments, 48.85% neutral comments and 27.60% negative comments. BanG Dream had 22.15% positive comments, 63.10% neutral comments and 14.75% negative comments.

Overall, both models suggest that BanG Dream comments were more neutral than Project SEKAI comments. However, the models differed in how they classified Project SEKAI sentiment. VADER classified Project SEKAI as more positive overall, while RoBERTa identified a higher proportion of negative comments. This suggests that Project SEKAI comments may be more emotionally complex or harder to classify, possibly due to fandom-specific language, sarcasm, humour or stronger emotional reactions.



This finding is important to the research question because it shows that two similar rhythm game fandoms can still have noticeable differences in emotional patterns on the same platform. BanG Dream appears more neutral across both sentiment models, while Project SEKAI appears more emotionally polarised.

c. Sentiment Model Agreement

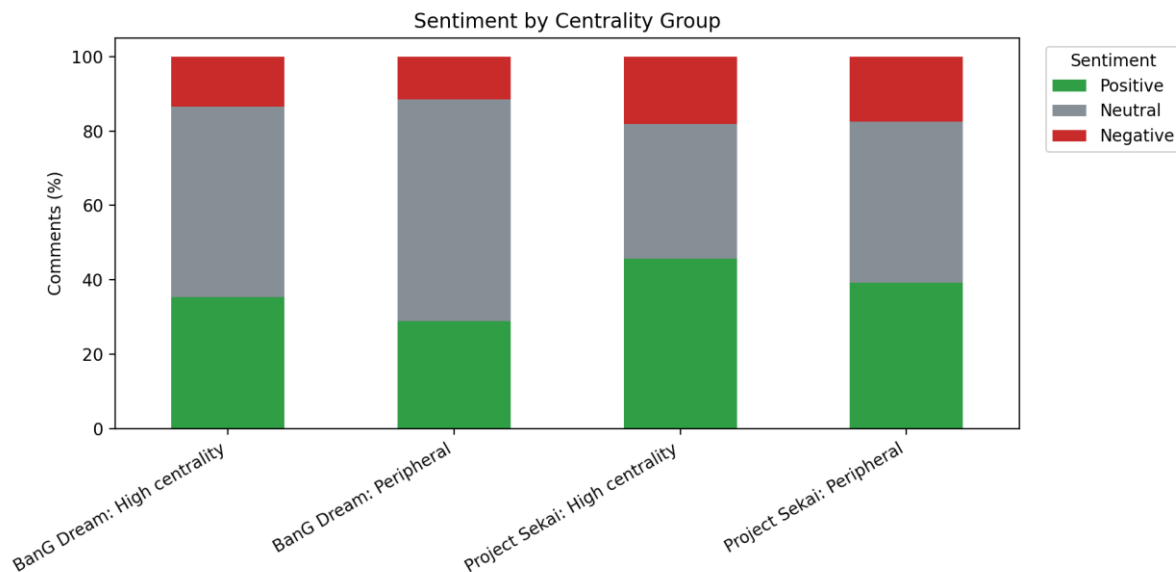


The VADER and RoBERTa results were also compared to examine whether both models classified comments in similar ways. Overall, the two models agreed on 65.26% of comments. However, the agreement rate differed between the two games. BanG Dream had a higher agreement rate of 73.18%, while Project SEKAI had a lower agreement rate of 57.49%.

This suggests that BanG Dream comments were easier for both models to classify consistently, likely because many comments were neutral or less emotionally intense. In contrast, the lower agreement rate for Project SEKAI suggests that its comments may contain more ambiguous, emotional, sarcastic, or fandom-specific language. This supports the earlier finding that Project SEKAI discussion appears more emotionally complex or polarised than BanG Dream discussion.

d. Network Structure Comparison

The reply networks for both communities were sparse and highly modular. This means that users did not interact evenly across the whole network. Rather, the interaction was divided into many separate groups or components.



In both communities, high centrality users had a higher percentage of positive comments than peripheral users. For Project SEKAI, 45.64% of high centrality user comments were positive, compared with 39.25% for peripheral users. For BanG Dream, 35.36% of high centrality user comments were positive, compared with 28.97% for peripheral users.

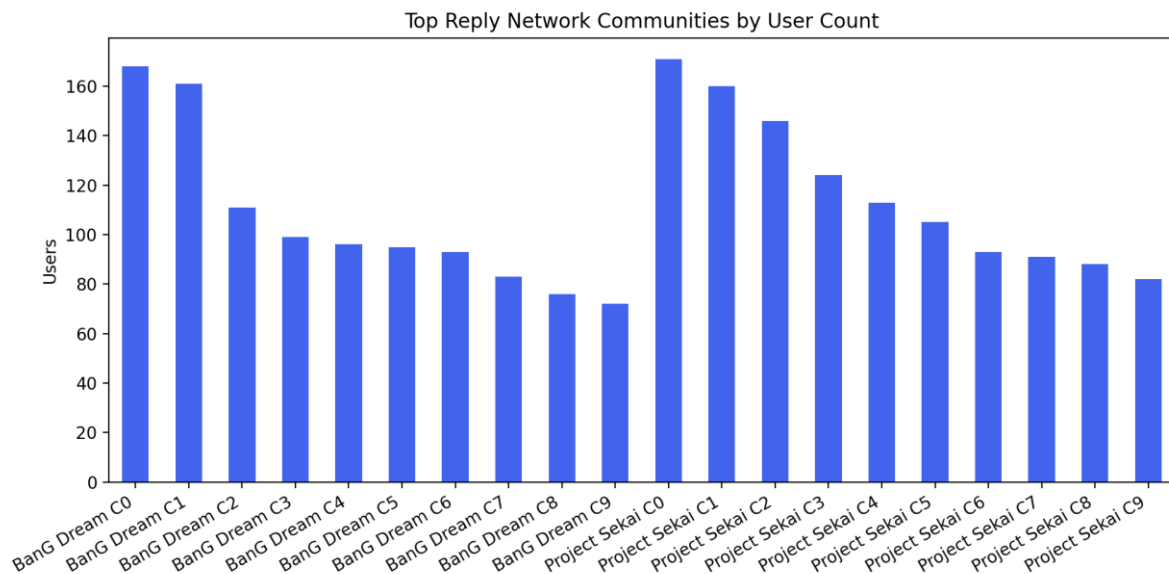
This suggests that more connected users in both fandoms may contribute more emotionally engaged or supportive comments. In contrast, high centrality users also had slightly higher negative percentages than their peripheral counterparts in both datasets. This suggests that central users may be more emotionally expressive overall, rather than only more positive.

Project SEKAI's high centrality users were especially positive compared with BanG Dream's high centrality users. Though, Project SEKAI also had a higher negative percentage. This supports earlier sentiment findings that Project SEKAI discussion appears more polarised as compared to BanG Dream's neutral discussion.

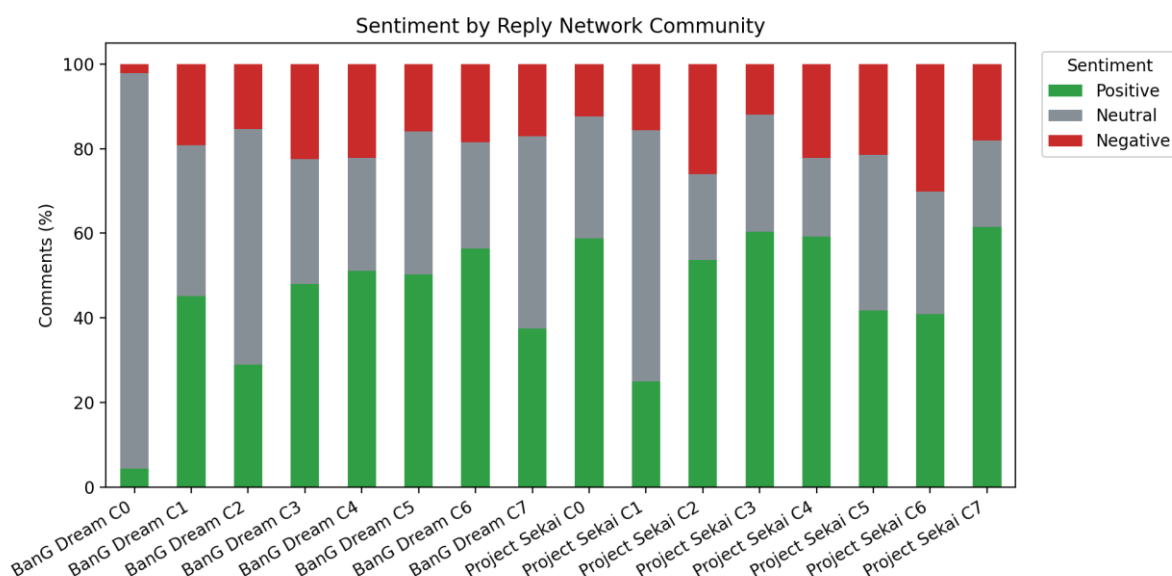
This result is useful because it connects network position with sentiment. It shows that the sentiment is not only different between games but also differs depending on user's role in the interaction network.

f. Community and Cluster Findings

Community detection found many clusters in both reply networks. Project SEKAI had 407 detected communities, while BanG Dream had 305. The high modularity scores suggests that these clusters are not random but meaningful structural divisions in the network.



The largest BanG Dream communities ranged from 168 users to 72 users in the top ten communities. While for the largest Project SEKAI communities, they were generally larger, matching its larger reply network overall. Many smaller communities in both networks contained only a few users, which suggests that a large part of the YouTube reply interaction happens within small, isolated conversations.



The community sentiment analysis showed that different clusters had different emotional patterns. For example, some BanG Dream communities were overwhelmingly neutral, while others had more positive or negative sentiment. In the community summary, one BanG Dream community had 93.55% neutral comments, while another had 56.31% positive comments. This shows that even within the same fandom, different interaction clusters can have different emotional tones.

This supports the idea that the YouTube rhythm game fandoms are not emotionally uniform. Instead, users appear to form smaller discussion clusters that may differ in both topic and sentiment. However, these clusters should not automatically be described as echo chambers. The analysis shows clustered interaction, but it does not prove that users are avoiding opposing views or reinforcing beliefs over time.

A more careful interpretation could be that both fandoms show evidence of self-contained interaction clusters. These clusters may become self-reinforcing if users mostly interact within the same group or around the same topic. Additionally, the use of longitudinal analysis would be needed to prove echo chamber behaviour more strongly.

g. Summary of Main Findings

The main findings from the results are:

- Both datasets are reasonably balanced, with 30 videos for each game with around 13000 comments per community.
- Project SEKAI had more unique users, replies and reply edges, thereby suggesting a more direct reply-based interaction in the sample.
- VADER classified Project SEKAI as having both higher positive and negative sentiment than BanG Dream.
- RoBERTa also showed Project SEKAI as less neutral and more negative than BanG Dream.
- BanG Dream was more neutral overall across both sentiment models.
- VADER and RoBERTa agreed more often on BanG Dream comments than on the same for Project SEKAI, suggesting Project SEKAI comments may be more emotionally complex or harder to classify.
- Both reply networks were sparse, fragmented and highly modular.
- Project SEKAI had a slightly larger and more modular network than BanG Dream.
- High centrality users were more positive than peripheral users in both communities under the VADER-based analysis.
- Different network communities showed different sentiment patterns, suggesting that the emotional tone varies across the different interaction clusters.

Overall, the results show that Project SEKAI and BanG Dream communities differ in both sentiment and network structure, even though they belong to the same broader rhythm game fandom category.

X. Discussion

Reiterating the research question:

How do interaction network structures and sentiment patterns differ between Project SEKAI and BanG Dream rhythm game communities on YouTube?

The results show that both communities form sparse and highly clustered YouTube reply networks. But they differ noticeably in their emotional tone and interaction scale. Although both communities belong to the same broader rhythm game category, the Project SEKAI dataset showed more reply-based interaction and more emotionally polarised sentiment, while the BanG Dream dataset showed a more neutral overall sentiment pattern.

a. Sentiment Differences Between the Communities

The sentiment results suggest that Project SEKAI discussion is more emotionally expressive and complex than BanG Dream discussion. Under VADER, Project SEKAI had both a higher positive and higher negative comment percentage, which means the community was not simply more positive or more negative, but more polarised overall.

The RoBERTa results supported the idea that Project SEKAI differs emotionally from BanG Dream, but in a more cautious way. RoBERTa classified fewer Project SEKAI comments as positive and more as negative compared with VADER. This difference suggests that the emotional tone of Project SEKAI comments may be more difficult to classify, especially if comments include sarcasm, memes, exaggeration, fandom-specific language or emotionally intense phrasing.

BanG Dream, in contrast, had a much higher neutral percentage across both sentiment models. This suggests that BanG Dream comments in the dataset were likely more descriptive, short, informational or less emotionally intense. This does not mean that the BanG Dream fandom is less engaged, but it may indicate that the sampled videos produced fewer strongly emotional reactions in the comments.

The model agreement results also support this interpretation. VADER and RoBERTa agreed more often on BanG Dream comments than on Project SEKAI comments. This suggests that BanG Dream comments were more straightforward for automated sentiment models to classify, while Project SEKAI comments were more ambiguous or emotionally varied.

This difference matters because it shows that two similar rhythm game fandoms can have different emotional cultures on the same platform. Even when the platform and broad genre are held constant, fandom-specific dynamics still appear to shape sentiment patterns.

b. Network Structure and Cluster Formation

Both communities formed sparse reply networks with very low density. This means users were not broadly connected to many others across the whole comment space. Instead, most users interacted with only a small number of others. This is typical of YouTube comment sections because many users post a single top-level comment or reply only once.

Both networks also showed very high modularity. Project SEKAI had a modularity score of 0.941378, while BanG Dream had a modularity score of 0.920611. These values suggest that both communities were strongly divided into clusters. In other words, users tended to interact within smaller groups rather than forming one evenly connected discussion network.

Project SEKAI's network was larger and slightly more modular than BanG Dream's. It had more nodes, edges, connected components, and detected communities. This suggests that Project SEKAI had more direct reply-based interaction but that this interaction was also divided into many separate clusters.

This supports the idea that rhythm game fandoms on YouTube can form self-contained interaction clusters. However, it is important not to overstate this as definite evidence of echo chambers. The analysis shows that clusters exist, but it does not prove that users are only exposed to similar opinions or that they avoid disagreement. Therefore, the findings are better described as evidence of clustered interaction patterns rather than full echo chambers.

c. Central Users and Sentiment

The centrality analysis showed that high-centrality users were more positive than peripheral users in both communities. This suggests that the users who are more connected in the reply network may contribute more positive or emotionally engaged comments.

For Project SEKAI, high-centrality users had 45.64% positive comments, compared to the 39.25% among peripheral users. For BanG Dream, the high centrality users had 35.26% positive comments compared with the 28.97% among peripheral users. This pattern suggests that central users may play a role in shaping the emotional tone of the discussions in the communities.

However, high-centrality users also had slightly higher negative percentages in both communities. This means that central users may simply be more expressive overall rather than purely more positive. They may be more likely to participate in debates, respond to others, defend opinions, or react strongly to content.

This is important because it connects the sentiment and network findings. The results suggest that sentiment is not evenly distributed across all users. Instead, a user's network position may relate to the type or intensity of sentiment they express.

d. Evidence of Self-Reinforcing Interaction Clusters

The high modularity scores and large number of detected communities suggest that both fandoms contain many interaction clusters. These clusters may be self-reinforcing if users mainly interact with others within the same cluster or around the same video topic.

In the context of YouTube, this clustering may happen because discussions are organised around specific videos. Users who comment on or reply within the same video may form temporary clusters around that content. If a video attracts users with similar opinions, humour, preferences or emotional reactions, the comment section may become a small self-reinforcing space.

However, the evidence is stronger for interaction clustering than for full echo chambers. Echo chambers usually imply repeated exposure to similar viewpoints and limited exposure to opposing views over time. This project uses a static dataset of comments and replies, so it cannot fully prove long-term reinforcement or isolation.

Therefore, the most accurate conclusion is that both Project SEKAI and BanG Dream YouTube communities show signs of clustered interaction structures, but further research would be needed to determine whether these clusters behave as true echo chambers.

e. Why the Two Communities Differ

The differences between the two communities may have been caused by several factors.

Firstly, the selected videos were not perfectly matched in content type. Project SEKAI had a variety of videos including more opinion-based, update based and discussion style videos, while BanG Dream videos included more official music, anime, live performances and gameplay related videos. Opinion based videos may naturally invite a more emotional or polarised comment section rather than official music or episode content.

Secondly, the two fandoms may have different audience cultures. Project SEKAI has a large and highly active online fandom, especially around characters, units, events and songs. This may encourage more debate and emotionally expressive commenting. BanG Dream, while also has a strong fandom, the sampled videos utilised may have attracted more general appreciation or neutral reactions instead.

Lastly, YouTube itself shapes interaction patterns. Unlike a discussion forum like Reddit, where long threaded responses are central to the platform, YouTube comment sections are primarily attached to the videos. This means that interaction clusters may form around specific video content rather than around stable community groups. As a result, clustering in this project may reflect video centred engagement, as much as fandom division.

f. Connection to the Research Question

Overall, the results show that Project SEKAI and BanG Dream YouTube communities differ in both sentiment patterns and interaction network structures.

Project SEKAI had a larger reply network, more direct reply interactions, more detected communities and a slightly higher modularity score. It also had a more polarised sentiment distribution, with both more positive and more negative comments than BanG Dream. This suggests that the Project SEKAI community in the dataset was more active in reply-based interaction and more emotionally expressive.

BanG Dream had a smaller reply network and a more neutral sentiment distribution. Its network was still highly modular, showing strong clustering, but its comments were less emotionally polarised overall.

Therefore, the findings suggest that both rhythm game communities form clustered interaction networks on YouTube, but the nature of those clusters and the sentiment patterns within them differ. Project SEKAI appears more emotionally intense and structurally fragmented, while BanG Dream appears more neutral but still clustered.

XI. Limitations

This study has several limitations. Firstly, sentiment analysis was performed using both VADER and RoBERTa, but neither model can perfectly interpret fandom-specific language. YouTube comments often include sarcasm, humour, emojis, slang, exaggeration, and game-specific references, which may lead to comments being classified differently across models. While RoBERTa provides more contextual understanding than VADER, the disagreement between models shows that sentiment labels should still be interpreted as approximate indicators rather than exact measures of emotional meaning.

Secondly, the network analysis was based only on YouTube reply interactions. This captures users replying to each other within comment threads, but does not capture broader social relationships such as likes, subscriptions, memberships, follower networks, or interactions across other social media platforms. Including additional platforms or richer forms of interaction data would provide a more complete picture of fandom community structure.

Moreover, YouTube reply networks are often sparse and one-directional, as many users comment only once or reply to a single thread. This can produce low-density networks with fragmented components. As a result, detected communities represent clusters of reply interaction, but they do not necessarily prove the existence of strict echo chambers in a broader social sense.

Finally, this study focuses only on two rhythm game communities on a single platform. The findings highlight differences between the sampled Project SEKAI and BanG Dream YouTube communities, but they should not be generalised to all rhythm game fandoms or to the same communities on other platforms.

XII. Conclusion

This project compared the YouTube communities of two major rhythm games, *Project SEKAI: Colorful Stage!* And *BanG Dream! Girls Band Party!*, using sentiment analysis and network analysis. The research question asked how interaction network structures and sentiment patterns differ between the two rhythm game communities on YouTube.

The results show that both communities form sparse, highly clustered reply networks. Both networks had very low density and high modularity, suggesting that user interaction is divided into many smaller clusters rather than one fully connected discussion space. This indicates that clustered interaction is a common feature of both rhythm game communities on YouTube.

However, the two communities differed in both scale and sentiment. Project SEKAI had more unique users, replies, reply edges and detected communities. The sentiment results also showed that Project SEKAI was more emotionally polarised than BanG Dream. VADER classified Project SEKAI as having both more positive and more negative comments, while RoBERTa classified it as less neutral and more negative than BanG Dream. BanG Dream had a higher proportion of neutral comments across both sentiment models, suggesting that its sampled YouTube discussion was less emotionally intense overall.

The centrality analysis also showed that high-centrality users were more positive than peripheral users in both communities. This suggests that more connected users may contribute more emotionally engaged or supportive comments. However, central users also showed slightly higher negative sentiment, meaning they may be more expressive overall rather than simply more positive.

Overall, the findings suggest that Project SEKAI and BanG Dream YouTube communities both form clustered interaction networks, but they differ in emotional tone and interaction intensity. Project SEKAI appears more active and emotionally polarised, while BanG Dream appears more neutral but still structurally clustered. The inclusion of both VADER and RoBERTa shows that the exact sentiment distribution depends on the model used, but both models support the broader conclusion that the two communities differ in emotional tone. These findings support the idea that even similar rhythm game fandoms on the same platform can develop different social and emotional patterns.

The project itself does not prove the existence of full echo chambers, but it does show evidence of self-contained interaction clusters. Future work could improve this analysis by using more closely matched video types, collecting data over time, comparing other relevant and active platforms, or analysing the topics discussed within each detected network community.

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